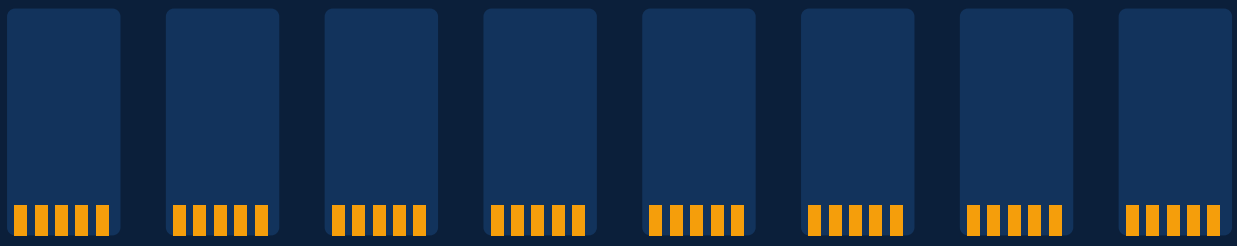

COMPUTER HARDWARE | ASSIGNMENT

Memory & Storage Fundamentals

RAM generations - HDD vs SSD vs NVMe - Hands-on installation tasks

TASK 1

DDR3 vs DDR4 vs DDR5 - Comparison Table

Each new DDR (Double Data Rate) generation roughly doubles the data rate of the previous one while lowering voltage to save power. They are **not** physically or electrically compatible - the notch position differs, so a DDR4 stick will not fit a DDR3 or DDR5 slot.

Feature	DDR3	DDR4	DDR5
Introduced	2007	2014	2020 (PCs from late 2021)
Speed (data rate)	800 - 2133 MT/s (1600 MT/s most common)	2133 - 3200 MT/s official; 3600+ with XMP	4800 - 8800 MT/s; 5600 - 6400 common in 2024+
Bandwidth per module (typical)	12.8 GB/s (DDR3-1600)	25.6 GB/s (DDR4-3200)	38.4 GB/s (DDR5-4800) and above
Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1 V
Pins (DIMM / laptop SO-DIMM)	240 / 204	288 / 260	288 / 262
Max capacity per module	Up to 16 GB (typically 2 - 8 GB)	Up to 64 GB	64 GB common; spec allows 512 GB
Key design features	Single 64-bit channel per module; basic design	Higher bank count; better efficiency than DDR3	Two 32-bit sub-channels per module; on-die ECC; power management chip (PMIC) on the module
Typical use cases	Legacy and old PCs, older laptops, retro builds, budget refurbished systems	Mainstream PCs and laptops (2015 - 2022), gaming, office work, many current budget builds	New gaming PCs, workstations, content creation, AI/ML, modern servers and latest laptops
Availability today	Mostly used/refurbished; being phased out	Widely available and affordable	Current standard; prices falling

Note: MT/s means mega-transfers per second (the true data rate). DDR4-3200 is often loosely called '3200 MHz', but its actual clock is 1600 MHz - data moves on both clock edges. Figures above are typical values; exact specs vary by manufacturer.

Summary

DDR3 is the old, slower, higher-voltage standard. **DDR4** is the long-running mainstream standard with a good balance of cost and speed. **DDR5** offers the highest speed, capacity and efficiency, making it the best choice for new, performance-focused systems. The motherboard decides which type you can use, so always check compatibility before buying RAM.

TASK 2

Locate, Photograph & Reinstall the RAM

Safety first: Shut down completely, unplug the charger/power cable, and for laptops remove the battery if it is removable. Hold the power button for 10 seconds to discharge. Touch bare metal on the case (or wear an anti-static strap) before handling parts. Handle RAM only by its edges. If unsure, follow your device manual or a trusted YouTube guide for your exact model.

Procedure

1	Power off the device and disconnect power. Place it on a clean, flat, non-carpeted surface.
2	Open the case: for a desktop remove the side panel; for a laptop open the service/RAM cover (or the bottom panel) using the correct screwdriver. Check your manual - some laptops have soldered RAM.
3	Locate the RAM slots (long, narrow slots next to the CPU on desktops; under a metal shield on laptops).
4	**Take a clear photo** of the installed RAM module showing its label (capacity, type and speed).
5	Remove the module: push the side clips outward (desktop) or the metal retaining clips (laptop). The stick pops up at an angle - pull it out gently by the edges.
6	Reinstall: align the notch on the module with the key in the slot, then press down firmly (desktop) or insert at ~30 degrees and press flat until the clips click (laptop).
7	Close the case, reconnect power and switch on. Confirm that the system boots normally.

Photo evidence

PASTE YOUR PHOTO HERE

Installed RAM module (label visible)

Verification record

RAM type (DDR3/4/5)	
Capacity (GB)	
Speed (MT/s)	
Slots total / used	
System booted OK? (Yes/No)	
Verified in Task Manager?	

- [ ] Power disconnected before opening
- [ ] Photo taken with label readable
- [ ] Module seated fully, clips locked
- [ ] System booted and RAM size detected correctly

TASK 3

HDD vs SSD vs NVMe - Key Differences

All three store data permanently, but they differ in the technology used and how the data travels to the CPU. Three key differences are shown below.

Key difference	HDD	SSD (SATA)	NVMe SSD
1. Speed & interface	About 80 - 160 MB/s over SATA; slow random access	About 500 - 550 MB/s over SATA III	About 3,500 - 7,000+ MB/s over PCIe (Gen 3 / 4 / 5); lowest latency
2. Technology & durability	Spinning magnetic platters and a moving read/write head; sensitive to shocks; audible noise	Flash memory chips; no moving parts; silent and shock-resistant	Flash memory using the NVMe protocol; no moving parts; may run hot and need a heatsink
3. Cost & capacity	Cheapest per GB; up to 20+ TB	Moderate price per GB; commonly 256 GB - 4 TB	Highest price per GB (falling); commonly 256 GB - 4 TB; M.2 stick form factor
Form factor	3.5 in or 2.5 in drive, SATA cable	2.5 in drive (or M.2 SATA)	M.2 slot (also PCIe cards)

Best-fit app / use case for each storage type

Storage	Everyday app / use case	Why it benefits most
HDD (capacity & cost)	Backing up WhatsApp media, photos and a large offline movie library played in VLC	Huge amounts of rarely-changed data are needed cheaply; speed is not critical.
SSD (everyday speed)	Windows start-up, Google Chrome with many tabs, MS Word / Excel	Fast boot and quick app launches and file opening without paying for top-end speed.
NVMe (maximum speed)	Video editing in CapCut / Premiere Pro, and loading large games (e.g. Valorant, BGMI emulator)	Very high read/write speeds cut rendering, scrubbing and game loading times.

Tip: Many people use both - an NVMe SSD for the operating system and active apps, plus an HDD for bulk storage. Replace the example apps above with ones you personally use if your instructor asks for your own daily apps.

TASK 4 Remove the Storage Drive & Identify the Connector

Constraint: Only do this on a device you own or have permission to open. If you cannot open one, use the YouTube-link option at the end of this task. Back up important data first, and power off / unplug as in Task 2.

Procedure

1	Shut down, unplug power, remove the battery if possible, and open the case or the storage cover.
2	Locate the drive: a 2.5 in / 3.5 in box with a SATA cable, or a small M.2 stick held by one screw.
3	Remove the screw or the cables/bracket and carefully take the drive out.
4	**Photograph the connector** and the label (SATA, M.2 SATA, or M.2 NVMe).
5	Reinstall the drive firmly (M.2: insert at an angle, press down, refit the screw) and close the case.
6	Power on and confirm the system detects the drive.

How to identify the connector

Type	What it looks like	Key / notch
SATA drive	2.5 in or 3.5 in case with two L-shaped connector segments (data + power)	L-shaped slots, no M.2 notch
M.2 SATA SSD	Small gumstick card (e.g. 2280 = 22 mm x 80 mm)	B+M key: two notches
M.2 NVMe SSD	Same gumstick size as M.2 SATA, often with a label naming NVMe / PCIe	M key: one notch (6 pins on one side)

PASTE YOUR PHOTO HERE

Connector type photo (SATA / M.2 SATA / M.2 NVMe)

Verification & record

Drive type (HDD / SSD / NVMe)	
Connector type (SATA / M.2)	
Brand & model	
Capacity	
Detected after reassembly? (Yes/No)	

How to confirm the drive is recognised

- [ ] Windows: right-click Start > Disk Management, or Task Manager > Performance > Disk
- [ ] Linux: run lsblk in a terminal

- [ ] BIOS/UEFI: check the storage list on the first screen
- [ ] System boots into the operating system normally

If you cannot open your device: YouTube option

Search YouTube for **'[your laptop/PC model] SSD replacement'** or **'[model] storage drive removal'**, pick a video for your exact model, and paste the link below.

Device brand & model	
YouTube video title	
YouTube link	

Conclusion: Knowing your RAM generation and storage type helps you choose compatible upgrades, troubleshoot confidently, and match hardware to the way you actually use your computer.